

Problem

The step response of an RLC circuit is given by

$$\frac{d^2 i}{dt^2} + 2 \frac{di}{dt} + 5i = 10$$

Given that $i(0) = 2$ and $di(0)/dt = 4$, solve for $i(t)$.

Step-by-step solution

Step 1 of 5

Given equation is,

$$\frac{d^2 i}{dt^2} + 2 \frac{di}{dt} + 5i = 10 \quad \dots (1)$$

The step response of the parallel RLC circuit is,

$$\frac{d^2 i}{dt^2} + \frac{1}{RC} \frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC} \quad \dots (2)$$

Compare equation (1) with equation (2).

$$\frac{1}{RC} = 2$$

$$\frac{1}{LC} = 5$$

$$\frac{I_s}{LC} = 10$$

Step 2 of 5

Determine the value of α .

$$\alpha = \frac{1}{2RC}$$

Replace 2 for $1/RC$ in the equation.

$$\alpha = \frac{2}{2}$$

$$= 1$$

Determine the value of ω_o .

$$\text{Consider } \omega_o = \frac{1}{\sqrt{LC}}$$

$$\omega_o = \frac{1}{\sqrt{LC}}$$

$$= \sqrt{5}$$

Step 3 of 5

Compare the values of α and ω_o .

By comparing, $\alpha < \omega_o$.

For $\alpha < \omega_o$ the circuit is under damped and the solution for under damped system is,

$$i(t) = I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \dots\dots (3)$$

Determine the value of I_s .

Consider,

$$\begin{aligned} \frac{I_s}{LC} &= 10 \\ I_s &= 10(LC) \\ &= 10\left(\frac{1}{5}\right) \\ &= 2 \end{aligned}$$

Determine the roots of equation (1).

$$\begin{aligned} s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_o^2} \\ &= -\alpha \pm j\omega_d \\ j\omega_d &= \sqrt{\alpha^2 - \omega_o^2} \\ &= \sqrt{1^2 - (\sqrt{5})^2} \\ &= j2 \\ \omega_d &= 2 \end{aligned}$$

Step 4 of 5

Consider that $i(0) = 2$.

Replace 0 for t in equation 2.

$$\begin{aligned} i(0) &= 2 + (A_1 \cos(0) + A_2 \sin(0)) e^{-\alpha(0)} \\ 2 &= 2 + A_1 \cos(0) \\ 2 &= 2 + A_1 \\ A_1 &= 0 \end{aligned}$$

Step 5 of 5

Consider $\frac{d}{dt} i(0) = 4$.

Differentiate equation 2 with respect to t .

$$\begin{aligned} \frac{di(t)}{dt} &= (-2A_1 \sin \omega_d t + 2A_2 \cos \omega_d t) e^{-t} - e^{-t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t) \\ \frac{di(0)}{dt} &= \cos [(-2A_1 \sin(0) + 2A_2 \cos(0)) e^{-0} - e^{-0} (A_1 \cos(0) + A_2 \sin(0))] \\ 4 &= 2A_2 - A_1 \\ 4 &= 2A_2 \\ A_2 &= 2 \end{aligned}$$

Replace 2 for I_s , 0 for A_1 , 2 for A_2 , and 2 for ω_d in equation 1.

$$i(t) = 2 + 2 \sin 2t e^{-t}$$

Therefore, the solution for $i(t)$ is $\boxed{2 + 2e^{-t} \sin 2t}$.